

```
import random
```

```
import pygame
```

```
# Terrain types
```

```
GRASS = 0
```

```
SAND = 1
```

```
WATER = 2
```

```
ROCK = 3
```

```
class Tile:
```

```
    def __init__(self, tile_type, size):
```

```
        self.tile_type = tile_type
```

```
        self.size = size
```

```
        if tile_type == GRASS:
```

```
            self.color = (50, 170, 70)
```

```
        elif tile_type == SAND:
```

```
            self.color = (220, 200, 140)
```

```
elif tile_type == WATER:
```

```
    self.color = (40, 120, 220)
```

```
elif tile_type == ROCK:
```

```
    self.color = (120, 120, 120)
```

```
def draw(self, screen, x, y, camera_x, camera_y):
```

```
    rect = pygame.Rect(
```

```
        x * self.size - camera_x,
```

```
        y * self.size - camera_y,
```

```
        self.size,
```

```
        self.size
```

```
    )
```

```
# Base terrain
```

```
pygame.draw.rect(
```

```
    screen,
```

```
    self.color,
```

```
    rect
```

```
)
```

```
# Small visual details
```

```
if self.tile_type == GRASS:
```

```
    for _ in range(3):
```

```
        px = rect.x + random.randint(3, self.size-3)
```

```
        py = rect.y + random.randint(3, self.size-3)
```

```
        pygame.draw.circle(
```

```
            screen,
```

```
            (30,130,50),
```

```
            (px,py),
```

```
            2
```

```
        )
```

```
elif self.tile_type == SAND:
```

```
    for _ in range(3):
```

```
        px = rect.x + random.randint(3, self.size-3)
```

```
        py = rect.y + random.randint(3, self.size-3)
```

```
        pygame.draw.circle(
```

```
    screen,  
    (190,170,110),  
    (px,py),  
    1  
)
```

```
elif self.tile_type == ROCK:
```

```
    pygame.draw.line(  
        screen,  
        (80,80,80),  
        (rect.x+5, rect.y+5),  
        (rect.x+self.size-5,  
        rect.y+self.size-5),  
        2  
    )
```

```
elif self.tile_type == WATER:
```

```
    pygame.draw.line(  
        screen,  
        (120,200,255),  
        (rect.x+5,  
        rect.y+self.size//2),
```

```
(rect.x+self.size-5,  
    rect.y+self.size//2),  
    2  
)
```

```
class World:
```

```
    def __init__(self, width=200, height=200, tile_size=29):
```

```
        self.width = width
```

```
        self.height = height
```

```
        self.tile_size = tile_size
```

```
        self.temperature= 25
```

```
        self.luminosity=80
```

```
        self.humidity=40
```

```
        self.terrain = "Grass"
```

```
        self.tiles = []
```

```
        # Visual obstacles
```

```
        self.obstacles = []
```

```
self.generate_world()
```

```
self.generate_obstacles()
```

```
def generate_world(self):
```

```
    for y in range(self.height):
```

```
        row = []
```

```
            for x in range(self.width):
```

```
                chance = random.random()
```

```
                if chance < 0.60:
```

```
                    terrain = GRASS
```

```
                elif chance < 0.75:
```

```
                    terrain = SAND
```

```
elif chance < 0.90:
```

```
    terrain = ROCK
```

```
else:
```

```
    terrain = WATER
```

```
row.append(
```

```
    Tile(
```

```
        terrain,
```

```
        self.tile_size
```

```
    )
```

```
)
```

```
self.tiles.append(row)
```

```
def generate_obstacles(self):
```

```
    for _ in range(50):
```

```
x = random.randint(  
    0,  
    self.width-1  
)
```

```
y = random.randint(  
    0,  
    self.height-1  
)
```

```
obstacle = pygame.Rect(  
    x*self.tile_size,  
    y*self.tile_size,  
    self.tile_size,  
    self.tile_size  
)
```

```
self.obstacles.append(obstacle)
```

```
self.obstacles.append(pygame.Rect(800,650,self.tile_size,self.tile_size))
```



```
# Future sensor use
```

```
def get_tile(self, x, y):
```

```
    if x < 0 or y < 0:
```

```
        return None
```

```
    if x >= self.width or y >= self.height:
```

```
        return None
```

```
    return self.tiles[y][x]
```

```
def draw(self, screen, camera_x, camera_y):
```

```
    # Draw terrain
```

```
    for y in range(self.height):
```

```
        for x in range(self.width):
```

```
self.tiles[y][x].draw(  
    screen,  
    x,  
    y,  
    camera_x,  
    camera_y  
)
```

```
# Draw obstacles
```

```
for obstacle in self.obstacles:
```

```
    obstacle_rect = pygame.Rect(  
        obstacle.x - camera_x,  
        obstacle.y - camera_y,  
        obstacle.width,  
        obstacle.height  
    )
```

```
    pygame.draw.rect(screen,(255,0,0),obstacle_rect)
```